

FikraCore Simulator - 90-Capability Priority Matrix

Priority: P0 = simulator foundation | P1 = operational/resilience | P2 = advanced intelligence & learning | P3 = optimization/planning | P4 = autonomous resilience

Rank	Capability	Priority	FCAPS	Mode	Primary Simulator Outcome
1	Hypothesis Falsification	P0	F/C/A/P/S	Reactive	Reject wrong explanations systematically
2	RCA Accuracy Benchmarking	P0	F	Reactive	Prove RCA against hidden truth
3	Failure Propagation Analysis	P0	F/P	Both	Reconstruct causal propagation
4	Cross-Domain Impact Analysis	P0	F/C/P/S	Both	Understand domain-to-domain effects
5	Service Impact Prediction	P0	F/C/A/P/S	Both	Map technical failures to services
6	Blast-Radius Prediction	P0	F/C/P/S	Both	Determine extent of impact
7	Model Contradiction Detection	P0	All	Both	Detect when reality contradicts FikraCore
8	Unknown-Unknown Discovery	P0	All	Both	Discover unanticipated knowledge/model gaps
9	Topology Gap Discovery	P0	F/C/A	Both	Identify missing nodes/edges/paths
10	Knowledge Gap Analysis	P0	All	Both	Identify what FikraCore does not know
11	Observability Gap Analysis	P0	F/P/S	Both	Identify missing evidence
12	Evidence Quality Analysis	P0	All	Reactive	Determine evidence usefulness
13	Evidence Trust Analysis	P0	All	Reactive	Detect unreliable/conflicting evidence
14	Decision Uncertainty Analysis	P0	All	Both	Quantify uncertainty instead of forcing RCA
15	Multi-Cause Failure Analysis	P0	All	Both	Avoid single-root assumption
16	Hidden Common Dependency Discovery	P0	All	Both	Discover shared upstream causes
17	Hidden Relationship Discovery	P0	All	Both	Infer candidate missing relationships
18	Counterfactual Analysis	P0	All	Both	Test what should happen if H is true
19	What-If Analysis	P0	All	Proactive	Explore failures before they occur
20	Vulnerability Discovery	P0	All	Proactive	Discover latent outage conditions
21	Next-Best-Evidence Analysis	P0	All	Reactive	Select evidence that reduces uncertainty fastest
22	Correlation Algorithm Benchmarking	P0	F	Reactive	Validate causal correlation approaches
23	Alarm Storm Simulation	P0	F	Reactive	Test reasoning under event floods
24	Fault Injection / Chaos Simulation	P0	F	Proactive	Controlled failure experimentation
25	Incident Replay	P0	F	Reactive	Reproduce known incidents deterministically
26	Monitoring Blind-Spot Discovery	P1	F/P/S	Both	Find failures monitoring cannot see
27	Dynamic Topology Discovery	P1	F/C/A	Both	Learn changing/transient paths
28	Knowledge Staleness Detection	P1	C/A	Both	Detect outdated FikraCore knowledge
29	Critical Failure Surface Analysis	P1	All	Proactive	Find smallest conditions causing major impact
30	Single Point of Failure Discovery	P1	F	Proactive	Find dangerous non-redundant dependencies
31	Minimal Cut-Set Analysis	P1	F/P	Proactive	Find minimum failure combinations

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32	N-1 / N-2 / N-k Failure Analysis	P1	F/P	Proactive	Validate resilience under combined loss
33	Cascading Failure Analysis	P1	F/P	Both	Discover cascading outage chains
34	Compound Risk Analysis	P1	All	Proactive	Analyze interacting risk factors
35	FCAPS Cross-Dimensional Risk Analysis	P1	All	Proactive	Find risks spanning FCAPS dimensions
36	Proactive Resilience Analysis	P1	All	Proactive	Identify weaknesses before incidents
37	Predictive Outage Analysis	P1	F/P	Proactive	Identify emerging outage conditions
38	Redundancy Validation	P1	F/A/P	Proactive	Verify redundancy actually works
39	Resilience Claim Validation	P1	F/A/P	Proactive	Test N+1 and architectural claims
40	Failover Validation	P1	F/A	Proactive	Verify failover behavior
41	Recovery Path Analysis	P1	F	Both	Understand restoration dependencies
42	Recovery Strategy Comparison	P1	F/C	Proactive	Compare remediation approaches
43	Recovery Time Prediction	P1	F	Both	Estimate restoration duration
44	Disaster Recovery Simulation	P1	F/C/S	Proactive	Validate major recovery scenarios
45	Change Risk Simulation	P1	C	Proactive	Predict CR-related risk
46	Pre-Change Impact Analysis	P1	C	Proactive	Test a CR before implementation
47	Change Collision Analysis	P1	C	Proactive	Detect interacting concurrent changes
48	Acceptance Risk Simulation	P1	A	Proactive	Test onboarding/release risks
49	Design Validation Before Deployment	P1	A/C	Proactive	Test designs before production
50	Performance Stress Simulation	P1	P	Proactive	Test behavior under load
51	Capacity Cliff Discovery	P1	P	Proactive	Find nonlinear performance collapse points
52	Capacity Exhaustion Forecasting	P1	P	Proactive	Predict resource exhaustion
53	Security-to-Service Impact Simulation	P1	S	Proactive	Translate security conditions into service impact
54	Customer Impact Prediction	P1	F/P	Both	Translate service impact to customers
55	Failure Domain Analysis	P1	F/A	Proactive	Identify shared failure boundaries
56	Alarm Coverage Analysis	P1	F	Both	Determine whether faults produce detectable signals
57	Alarm Suppression Validation	P1	F	Both	Ensure suppression does not hide important faults
58	Investigation Path Optimization	P1	F	Reactive	Find efficient diagnostic sequence
59	Mean-Time-to-Isolate Optimization	P1	F	Reactive	Reduce fault-isolation time
60	Mean-Time-to-Understand Optimization	P1	F	Reactive	Reduce time to causal understanding
61	Mean-Time-to-Recover Optimization	P1	F	Both	Improve restoration performance
62	Runbook Validation	P1	F/C/A/P/S	Both	Test whether runbooks actually work
63	Operational Procedure Validation	P1	All	Both	Validate domain-engineer workflows
64	Historical Incident Mutation	P2	F	Both	Turn known incidents into unseen variants
65	Scenario Mutation Testing	P2	All	Both	Prevent memorization of fixed scenarios
66	Synthetic Incident Generation	P2	All	Both	Generate controlled new incidents

Rank	Capability	Priority	FCAPS	Mode	Primary Simulator Outcome
67	Novel Failure Generation	P2	All	Proactive	Test unseen failure mechanisms
68	Adversarial Scenario Generation	P2	All	Both	Deliberately challenge assumptions
69	Monte Carlo Outage Simulation	P2	All	Proactive	Explore large failure spaces
70	Digital Twin Experimentation	P2	All	Proactive	Experiment safely on network representation
71	Agent Reasoning Benchmarking	P2	All	Both	Measure reasoning quality
72	Agent Confidence Calibration	P2	All	Both	Align confidence with correctness
73	Agent Learning Validation	P2	All	Both	Prove learning improves future runs
74	Human-vs-Agent Decision Analysis	P2	All	Both	Compare reasoning and identify gaps
75	HITL Effectiveness Analysis	P2	All	Both	Measure where human intervention adds value
76	Shift-Handover Resilience Testing	P2	All	Reactive	Validate 24x7 operational continuity
77	Skill Readiness Assessment	P2	All	Both	Measure AI-first to AI-native readiness
78	Autonomous Action Readiness	P2	All	Both	Decide which actions may safely automate
79	Service Criticality Analysis	P2	All	Proactive	Rank services by operational consequence
80	Dependency Centrality Analysis	P2	All	Proactive	Identify highly influential dependencies
81	Risk Hotspot Mapping	P2	All	Proactive	Visualize concentrated network risk
82	Resilience Scoring	P2	All	Proactive	Quantify resilience by service/domain
83	Service Survivability Analysis	P2	F/P	Proactive	Measure service survival under progressive failure
84	Network Fragility Analysis	P2	All	Proactive	Identify structurally fragile areas
85	Network Robustness Analysis	P2	All	Proactive	Quantify resistance to disturbances
86	Preventive Action Prioritization	P3	All	Proactive	Rank mitigations by expected risk reduction
87	Investment / Capacity Upgrade Prioritization	P3	C/A/P	Proactive	Convert simulation into CAPEX priorities
88	Architecture Alternative Comparison	P3	All	Proactive	Compare competing designs
89	Future Network Architecture Simulation	P3	All	Proactive	Test future network designs
90	Autonomous Resilience Experimentation	P4	All	Autonomous	AI discovers what should be tested next

Priority Roadmap

P0 - Understand and challenge reality (1-25)

Prove causal reasoning first: imperfect evidence, cross-domain correlation, hypotheses, falsification, model contradiction, topology/knowledge discovery, RCA, what-if foundations and hidden-ground-truth benchmarking.

P1 - Break the network safely before reality does (26-63)

Use proactive simulation across Fault, Change, Acceptance, Performance and Security: SPOF/N-k, cascades, resilience claims, change risk, capacity stress, failover, recovery, monitoring coverage and operational procedure validation.

P2 - Learn and measure resilience (64-85)

Generate scenario variants and novel failures, benchmark agent reasoning, calibrate confidence, validate learning and HITL, and derive survivability, fragility, robustness and resilience scores.

P3 - Turn discovery into network decisions (86-89)

Prioritize preventive actions and investment, compare architecture alternatives and evaluate future network designs before production.

P4 - Autonomous Resilience Experimentation (90)

FikraCore identifies an uncertain assumption, asks what happens if it fails, designs and runs a safe experiment, discovers weaknesses, proposes mitigation, re-simulates and learns only after validation.

Long-term progression: RCA of what happened -> resilience analysis of what could happen -> autonomous discovery of what nobody thought to test.